

Mapping for the Environment – Online Mapping Tool Prototype

Abstract

Do you also think nature needs a good lawyer?

Clean water and a healthy environment need legal protection—help us build strong cases.

Develop a powerful web-based mapping prototype that turns public environmental data into trusted evidence for legal action. The user-friendly tool **will be used** in **real-world** practice. It will enable non-technical users to create professional, reliable, and data-driven environmental documentation.

Develop an **interactive mapping platform** that allows users to **visualise, combine, and export** complex Dutch **public datasets**, such as those from PDOK.nl and the Bestrijdingsmiddelenatlas. The tool will generate **layer-based maps** and **data exports** that can be used directly in **environmental court cases**.

Work closely with **driven environmental lawyers** and **senior engineers** from 010 Coding Collective, who will provide technical guidance and real-world experience.

The project is commissioned by **Stichting MOB** and **Lawyers for the Earth (Advocaat van de Aarde)**, renowned organisations dedicated to defending nature, water, and public health through environmental law.

Background

There is a growing need for an accessible online mapping tool that enables users to explore spatial information related to the environment, land use, and living conditions. By combining multiple map layers and data sources, relationships can be made visible and data can easily be reused for analysis and reporting.

Project Objective

The objective of this project is to develop a working prototype of an online tool that allows users to:

- switch map layers on and off;
- combine and filter map layers;
- create new layers (for example, a buffer layer with an x-km radius around a Natura 2000 area);
- download underlying data (Excel, odf);
- export the map view (e.g., as an image or PDF);
- export the credentials of the sources used.

The prototype must be able to use data available via **PDOK.nl** and must be prepared for integration of additional (public) datasets.

Functional Scope

1. General Functionalities

- Interactive, web-based map.
- Layer management (on/off, order, transparency).
- Selection of map objects (by clicking and/or area selection).

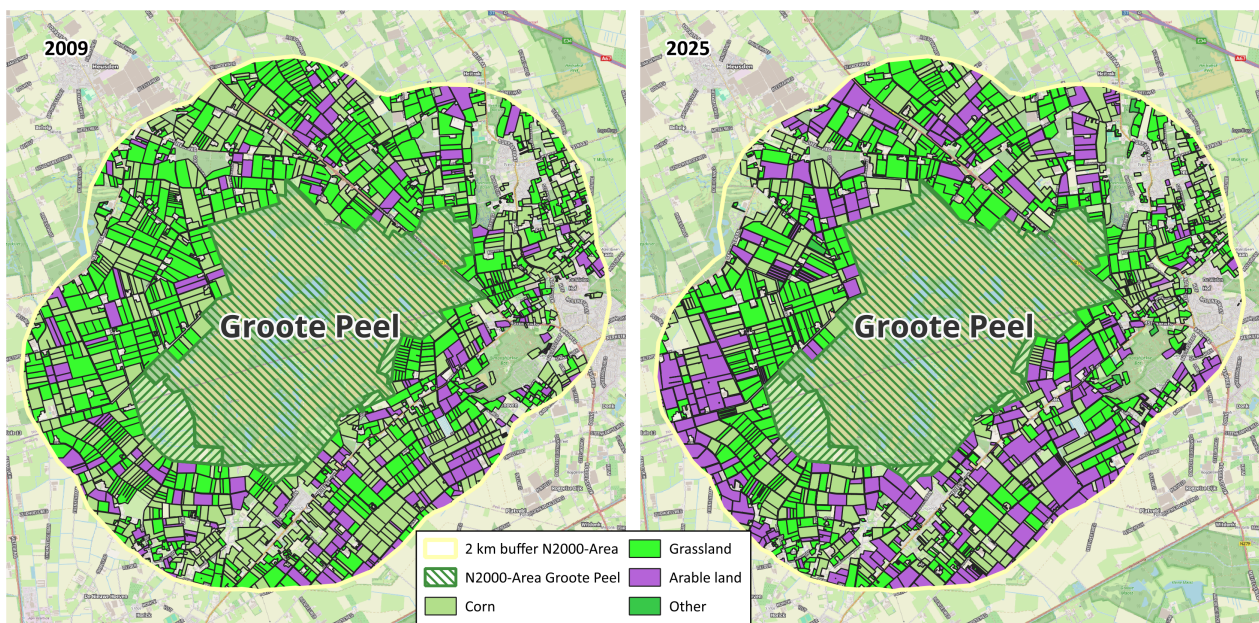
- Download of selected and/or filtered data to Excel/open office.
- Export of the current map view.
- Export the credentials of the sources used.

2. Map Layers (PDOK and Other Sources)

Initially, students may focus on data available through **PDOK.nl**. If further scoping is required, the clients will indicate which datasets must be included. At minimum, the following datasets are essential:

a. BRP Crop Parcels/BRP Gewaspercelen (including historical data)

- Display of crop parcels per year.
- Ability to select specific crops.
- Ability to group multiple crops into one category.

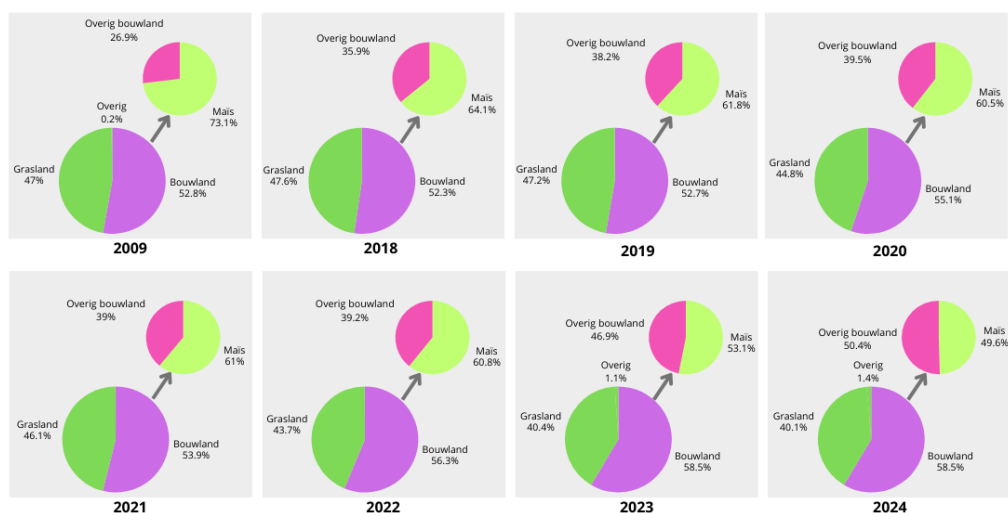


- Selection of parcels within a defined area (e.g., municipality, province, within an x-km radius of a Natura 2000 area, or manual selection).
- Calculation of parcel surface area.
- Pivot tables showing crops and corresponding areas within a selected region, with export to Excel for further analysis.

Example: Analyses area 250 meter within N2000-Area Groote Peel (2024)

Crop	Arable land	Grassland	Landscape element	Nature parcels	Total Area (ha)
Aardappelen, consumptie	30,82				30,82
Aardbeien open grond, vermeerdering	0,22				0,22
Agrarisch natuurmengsel		3,83			3,83
Bieten, suiker-	22,25				22,25
Bladrammenas	2,59				2,59
Boomgroep				0,05	0,05
Bosje				0,37	0,37
Erwten, groene/gele (groen te oogsten)	4,97				4,97
Gladiool, bloembollen en - knollen	5,52				5,52
Granen, overig	2,27				2,27
Grasland, blijvend		69,83			69,83
Grasland, natuurlijk. Met landbouwactiviteiten.		85,01			85,01
Grasland, tijdelijk		95,63			95,63
Groene braak, spontane opkomst	1,41				1,41
Houtwal en houtsingel				1,05	1,05
Italiaans raaigras, groenbemesting, vanggewas	0,08				0,08
Kerstbomen	1,73				1,73
Luzerne	3,07				3,07
Mais, corncob mix	5,34				5,34
Mais, korrel-	1,48				1,48
Mais, snij-	53,88				53,88
Maiskolvensilage	1,59				1,59
Natuurterreinen (incl. heide)				13,13	13,13
Overige gras, groenbemesting, vanggewas	0,32				0,32
Poel en klein historisch water				0,32	0,32
Sierconiferen, open grond	1,91				1,91
Sierheesters en klimplanten, open grond	1,19				1,19
Sloot				2,39	2,39
Stroken wild gras				0,28	0,28
Struweelrand				0,02	0,02
Tarwe, winter-	3,76				3,76
Triticale	3,29				3,29
Uien, gele zaai-	7,51				7,51
Water, overig				0,01	0,01
Totaal Resultaat	155,20	254,30	4,48	13,13	427,11

Development agricultural parcels 2 km surrounding N2000-area De Groote Peel



Data: Basisregistratie Gewaspercelen (Ministerie van Economische Zaken) te downloaden via pdok.nl

Kadaster#	Opp [Ha]	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	
MELO0-D-00	0,943	Mais	Mais	Mais	Mais	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Aardappelen	
MELO0-D-105	0,477	Mais	Mais	Mais	Mais	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Aardappelen	
MELO0-D-106	0,487	Mais	Mais	Mais	Mais	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Aardappelen	
MELO0-D-1855	1,447	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Aardappelen	
MELO0-D-1880	1,856	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Aardappelen	
MELO0-D-2078	0,321	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Aardappelen	
MELO0-D-2079	0,481	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Ervten	Wortels	Aardappelen	
MELO0-D-2167	0,631	Aardappelen	Mais	Groenten	Grasland	Grasland	Grasland	Grasland	Grasland	Mais	Grasland	Grasland	Grasland	Grasland	Mais	Gladiol	Ervten	Aardappelen	
MELO0-D-2215	0,044	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Mais	Grasland	Grasland	Grasland	Grasland	Aardappelen	Grasland	Ervten	Aardappelen	
MELO0-D-2285	0,317	Mais	Mais	Mais	Mais	Mais	Mais	Mais	Mais	Mais	Mais	Mais	Aardappelen	Mais	Bieten	Ervten	Wortels	Aardappelen	
MELO0-D-2289	0,961	Mais	Mais	Mais	Mais	Mais	Mais	Mais	Mais	Grasland	Grasland	Grasland	Grasland	Grasland	Mais	Ervten	Wortels	Aardappelen	
MELO0-D-2324	1,742	Aardappelen	Mais	Groenten	Grasland	Grasland	Grasland	Grasland	Grasland	Mais	Grasland	Grasland	Grasland	Grasland	Mais	Ervten	Gladiol	Ervten	Aardappelen
MELO0-D-316	1,034	Grasland	Mais	Aardappelen	Grasland	Mais	Mais	Grasland	Grasland	Grasland	Grasland	Mais	Wortels	Grasland	Grasland	Mais	Mais	Aardappelen	
MELO0-D-318	1,630	Mais	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Mais	Mais	Mais	Aardappelen	
NDW00-N-106	2,011	Mais	Mais	Mais	Mais	Mais	Mais	Mais	Mais	Mais	Mais	Mais	Mais	Mais	Mais	Mais	Mais	Aardappelen	
NDW00-N-107	1,100	Grasland	Grasland	Grasland	Grasland	Mais	Mais	Mais	Mais	Mais	Mais	Mais	Mais	Mais	Mais	Mais	Mais	Aardappelen	
NDW00-N-108	0,818	Grasland	Grasland	Grasland	Grasland	Mais	Mais	Mais	Mais	Mais	Mais	Mais	Mais	Mais	Mais	Mais	Mais	Aardappelen	
NDW00-N-159	1,503	Grasland	Grasland	Mais	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Mais	Mais	Mais	Mais	Mais	Mais	Mais	Aardappelen	
NDW00-N-110	2,695	Mais	Mais	Mais	Mais	Mais	Mais	Mais	Mais	Mais	Tarwe	Mais	Mais	Mais	Mais	Mais	Mais	Aardappelen	
NDW00-N-1281	2,231	Mais	Mais	Mais	Mais	Mais	Mais	Mais	Mais	Mais	Mais	Mais	Mais	Mais	Mais	Mais	Mais	Aardappelen	
NDW00-N-1283	2,648	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Mais	Mais	Aardappelen	
NDW00-N-1284	0,119	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Mais	Mais	Aardappelen	
NDW00-N-1285	1,597	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Mais	Mais	Mais	Mais	Mais	Mais	Mais	Aardappelen	
NDW00-N-1287	2,766	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Mais	Mais	Aardappelen	
NDW00-N-1288	0,135	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Mais	Mais	Aardappelen	
NDW00-N-1293	2,804	Mais	Mais	Mais	Mais	Mais	Mais	Mais	Mais	Aardappelen	Mais	Mais	Mais	Aardappelen	Mais	Bieten	Mais	Aardappelen	
NDW00-N-240	2,799	Mais	Mais	Mais	Mais	Aardappelen	Mais	Mais	Mais	Aardappelen	Mais	Bieten	Mais	Aardappelen	Chicoerei	Mais	Grasland	Aardappelen	
NDW00-N-7	2,919	Tarwe	Aardappelen	Grasland	Bieten	Aardappelen	Tarwe	Mais	Bieten	Aardappelen	Mais	Tarwe	Aardappelen	Tarwe	Tarwe	Ervten	Bieten	Aardappelen	
NDW00-O-1128	2,100	Mais	Mais	Mais	Mais	Mais	Mais	Mais	Mais	Uien	Aardappelen	Mais	Mais	Mais	Mais	Mais	Mais	Aardappelen	
NDW00-S-301	2,559	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Mais	Mais	Mais	Aardappelen	Mais	Mais	Mais	Grasland	Aardappelen
NDW00-O-327	0,209	Grasland	Mais	Aardappelen	Grasland	Mais	Mais	Grasland	Grasland	Grasland	Grasland	Mais	Wortels	Grasland	Grasland	Mais	Mais	Aardappelen	
NDW00-O-328	2,815	Grasland	Mais	Aardappelen	Grasland	Mais	Mais	Grasland	Grasland	Grasland	Grasland	Mais	Wortels	Grasland	Grasland	Mais	Mais	Aardappelen	
NDW00-S-335	1,579	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Mais	Grasland	Grasland	Grasland	Grasland	Aardappelen	
NDW00-O-465	1,630	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Mais	Mais	Mais	Aardappelen	Mais	Mais	Mais	Grasland	Aardappelen
NDW00-O-466	1,630	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Mais	Mais	Aardappelen	Mais	Mais	Mais	Grasland	Aardappelen
NDW00-O-617	1,100	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Grasland	Mais	Mais	Mais	Aardappelen	Mais	Mais	Mais	Grasland	Aardappelen
NDW00-O-997	4,370	Hazelzoden	Hazelzoden	Notenbomen	Notenbomen	Notenbomen	Notenbomen	Notenbomen	Notenbomen	Notenbomen	Notenbomen	Notenbomen	Notenbomen	Notenbomen	Notenbomen	Mais	Mais	Mais	Aardappelen

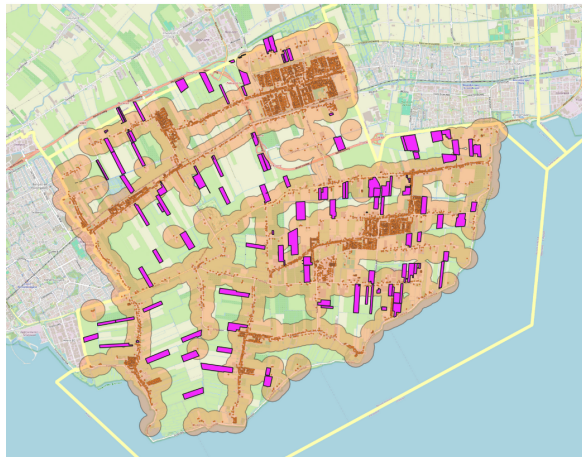
c. Natura 2000 Areas

- Display of Natura 2000 areas.
- Display of calculated buffers around these areas (configurable distance).

d. BAG (Basic Registration of Addresses and Buildings)

- Display of usage locations/buildings.

DRECHTERLAND - FLORICULTURE



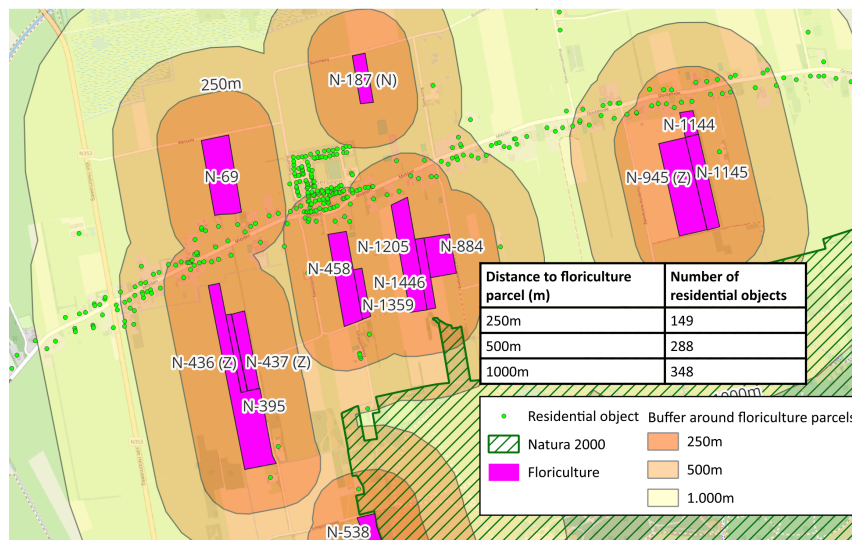
395 HECTARES

Floriculture

204 HECTARES

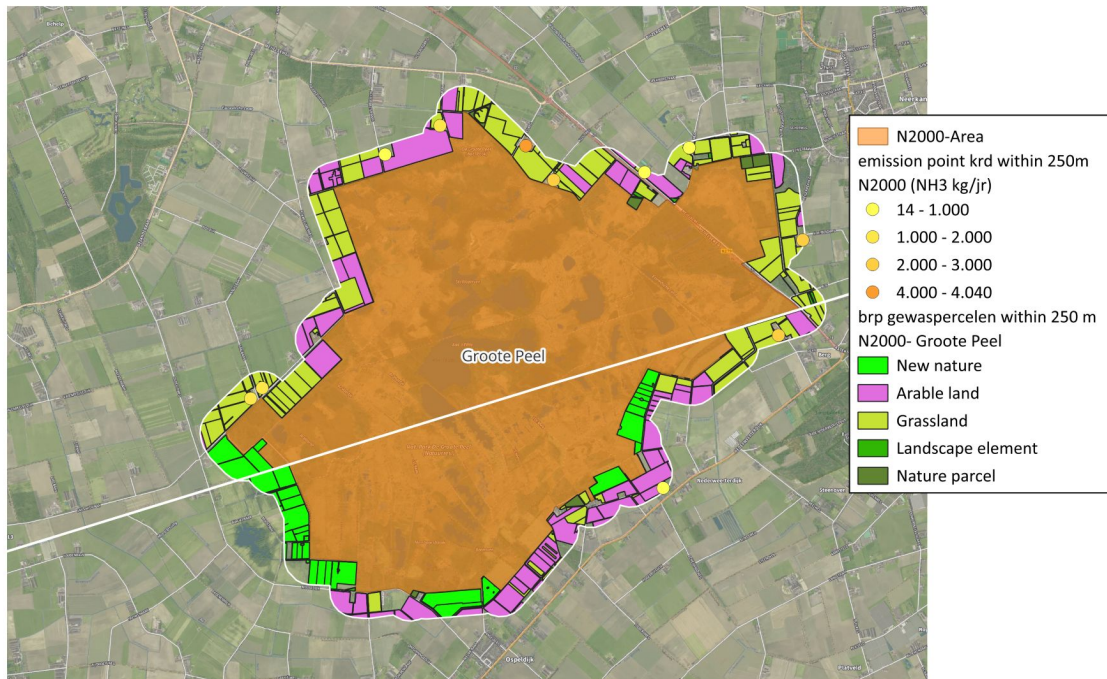
within 250 meter of a
residence location

- Relationship with surrounding parcels and areas.
- Counting of residential/usage locations within a defined buffer.



e. Emission Data for Animal Housing

- Emission figures for livestock facilities, as published via <https://krd.igoview.nl/> or released through WOO (Freedom of Information) requests.



- (The illustration includes data not derived from open sources: “New Nature within 250 m of Natura 2000 area”.)

f. Schools and Healthcare Institutions

- Point layers showing locations of schools and healthcare facilities.
- (These datasets are not yet available to the project team and may need to be sourced from public providers.)

3. External / Additional Datasets (Not Yet Implemented)

These datasets do not need to be fully implemented but must be considered in the system design:

g. Pesticides Atlas

<https://www.bestrijdingsmiddelenatlas.nl/atlas/1/1>

- Relevant measurement data included in the atlas.
- Linking agricultural parcels with detected pesticides.

h. Water Quality Portal

<https://www.waterkwaliteitsportaal.nl/>

- The client is developing a separate tool to access data from this portal.
- The system design must allow future integration of this data.

Technical and Design Principles

- Use of open data (including PDOK).
- Transparent data provenance (source attribution).

- Modular design, allowing extension with additional layers.
- Strong focus on usability and clear visualization.

Results / Deliverables

- A working online prototype of the mapping tool.
- A short technical documentation (architecture, data sources, maintenance).
- Overview of datasets used and their limitations.
- Recommendations for further development.

Scope and Limitations

- The prototype must function at minimum with all datasets provided through PDOK.
- The absolute minimum requirement is full functionality for layers a through f.
- If further limitations are required, necessary steps for integration must be described.
- Legal assessment of data usage is limited to basic license verification.

Current Focus

The current focus is on integrating and visualizing agricultural, environmental, and living environment data within a single, coherent mapping platform, with emphasis on exploration, analysis, and export functionality.

Input from the Clients

Throughout the project, students will receive periodic feedback and participate in brainstorming sessions with driven legal professionals. Basic expertise on off-line spacial analyses is available for knowledge sharing. The experts will provide feedback on the practical usability of the tool and indicate which analyses are useful and necessary for legal proceedings.

In addition, experienced IT professionals from 010 Coding Collective will actively contribute to the project. They will support the team in designing a robust and scalable prototype that can be further developed, expanded, and sustainably maintained after delivery.

Clients

Stichting MOB (Mobilisation for the Environment) is a Dutch environmental organization that works to protect nature and biodiversity by monitoring environmental compliance and initiating legal action against harmful practices. The organization focuses on issues such as nitrogen pollution, pesticide use, pollution of water and the enforcement of environmental regulations. <https://stichtingmobilisation.nl/>

Lawyers for the Earth (Advocaat van de Aarde) is an independent, volunteer-based legal organization that supports citizens and civil society in defending their right to a healthy environment. It uses strategic litigation and legal expertise to hold polluters accountable and to strengthen environmental protection through the rule of law. <https://advocaatvandeearde.nl/>